

Introduction to Data Science
Final Project Presentation

Traffic Volume Prediction Using Atmospheric Reanalysis Data

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Outline

- Introduction and Scope
- Data collection
- Data exploration
- Model selection
- Results
- Discussion
- Conclusion

Introduction and Scope

- Aim: Evaluate the feasibility of using atmospheric reanalysis data (estimated historical global climate data) to predict traffic volume, in contrast to local measurement of weather conditions.



<https://wmo.int/media/news/wmo-issues-new-guidelines-evaluation-of-weather-and-climate-extremes>



<https://motorlease.com/article/traffic-jams-explained/>

Introduction and Scope

- Scope of this project
 1. Gather traffic data with local weather measurements
 2. Gather weather data from atmospheric reanalysis dataset at place of measurement
 3. Build regression models to predict traffic volume based on both weather data
 4. Compare goodness of fit between the weather data acquisition methods used!

Data collection

- Metro Interstate Traffic Volume (**ATR***)
 - “Hourly weather features and holidays included for impacts on traffic volume”

 **Metro Interstate Traffic Volume**
Donated on 5/6/2019

Hourly Minneapolis-St Paul, MN traffic volume for westbound I-94. Includes weather and holiday features from 2012-2018.

Dataset Characteristics	Subject Area	Associated Tasks
Multivariate, Sequential, Time-Series	Other	Regression
Feature Type	# Instances	# Features
Integer, Real	48204	8

Dataset Information ^

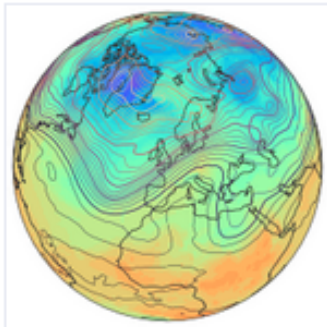
Additional Information
Hourly Interstate 94 Westbound traffic volume for MN DoT ATR station 301, roughly midway between Minneapolis and St Paul, MN. Hourly weather features and holidays included for impacts on traffic volume.

*Automatic Traffic Recorder

<https://archive.ics.uci.edu/dataset/492/metro+interstate+traffic+volume>

Data collection

- ERA5 hourly data on single levels from 1940 to present (**ERA5**)
 - Global climate and weather reanalysis dataset



⚠ The final validated ERA5 differs from ERA5T from July 2024 until further notice - please refer to our [Forum announcement](#) for details and watch it for further updates on this.

ERA5 hourly data on pressure levels from 1940 to present

Updated today

ERA5 is the fifth generation ECMWF reanalysis for the global climate and weather for the past 8 decades. Data is available from 1940 onwards. ERA5 replaces the ERA-Interim reanalysis. Reanalysis combines model data with observations from across the world into a globally complete and consistent dataset...

Reanalysis

Copernicus C3S

Global

Past

Atmosphere (surface)

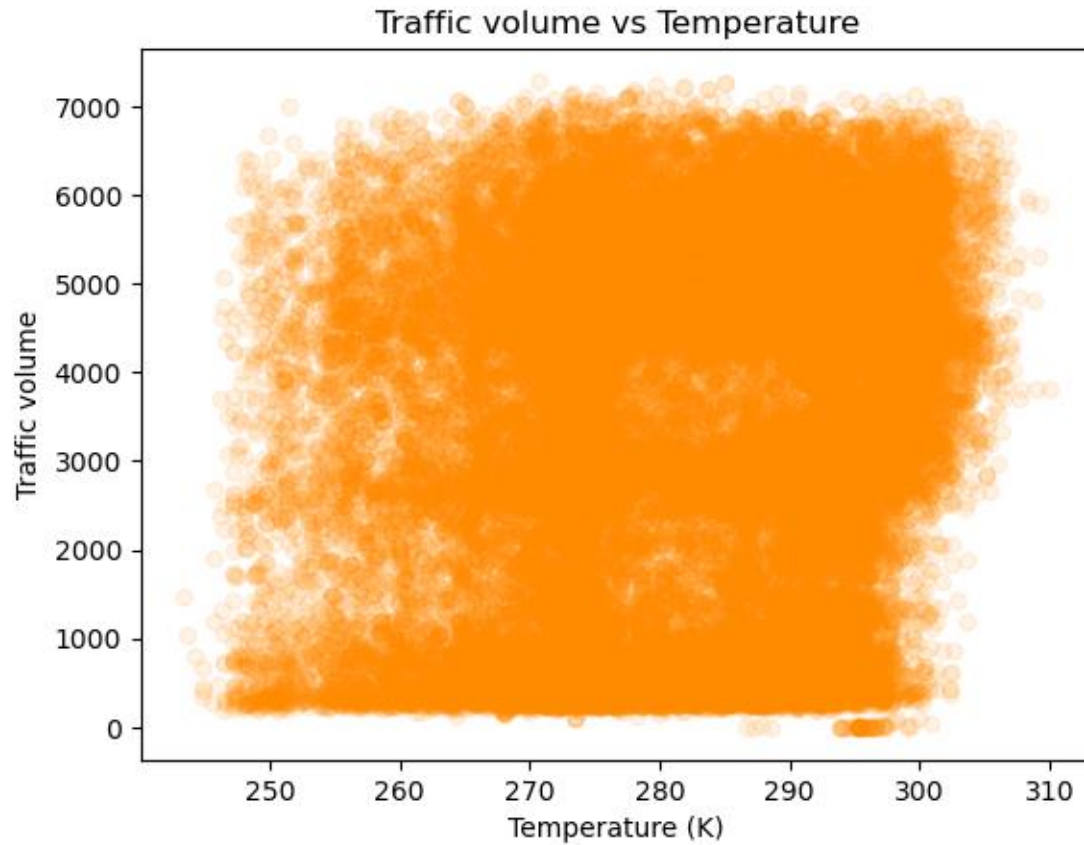
Atmosphere (upper air)

<https://cds.climate.copernicus.eu/datasets/reanalysis-era5-single-levels>

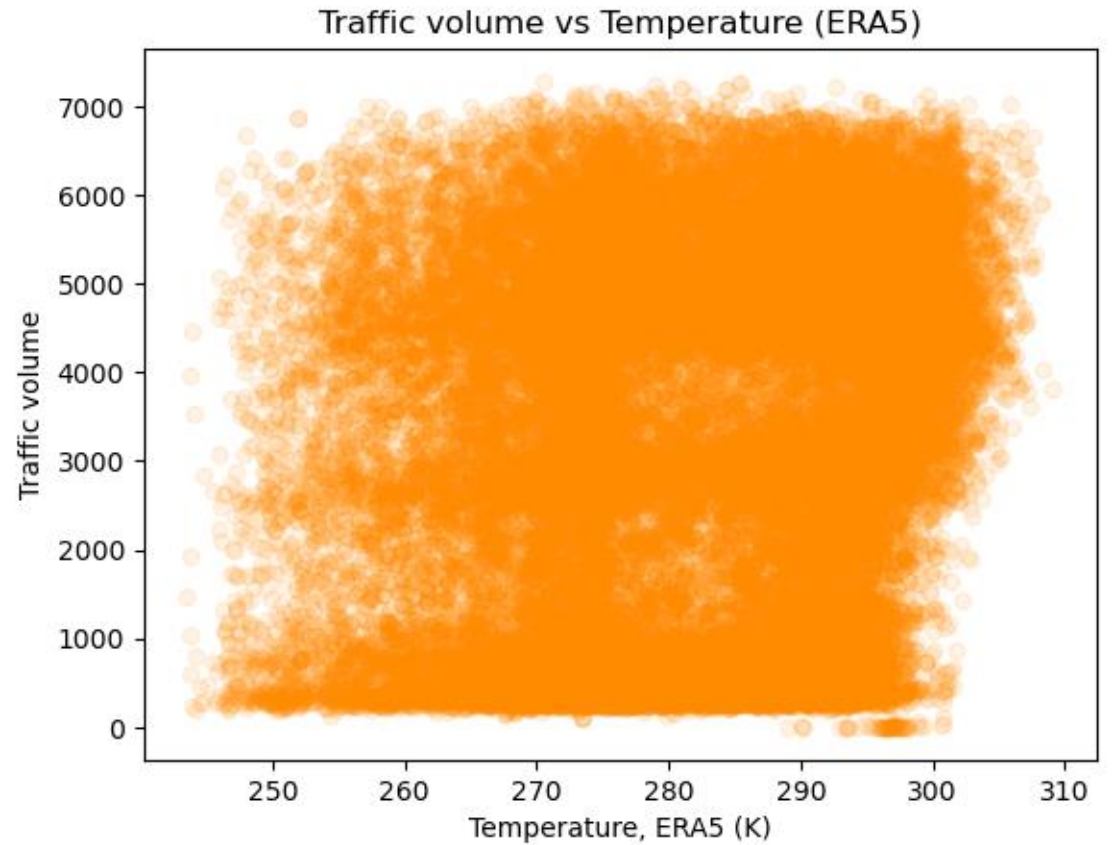
Data exploration

- ATR columns
 - Local date and time (CST)
 - Temperature
 - 1h rainfall (mm)
 - 1h snowfall (mm)
 - Cloud cover
 - Holiday
 - Weather description (Main)
 - Weather description (Detailed)
 - **Traffic volume**
- ERA5 columns
 - Date and time (UTC)
 - 2m temperature
 - Total precipitation (m)
 - Snowfall (m)
 - Cloud cover
 - Precipitation type

Data exploration



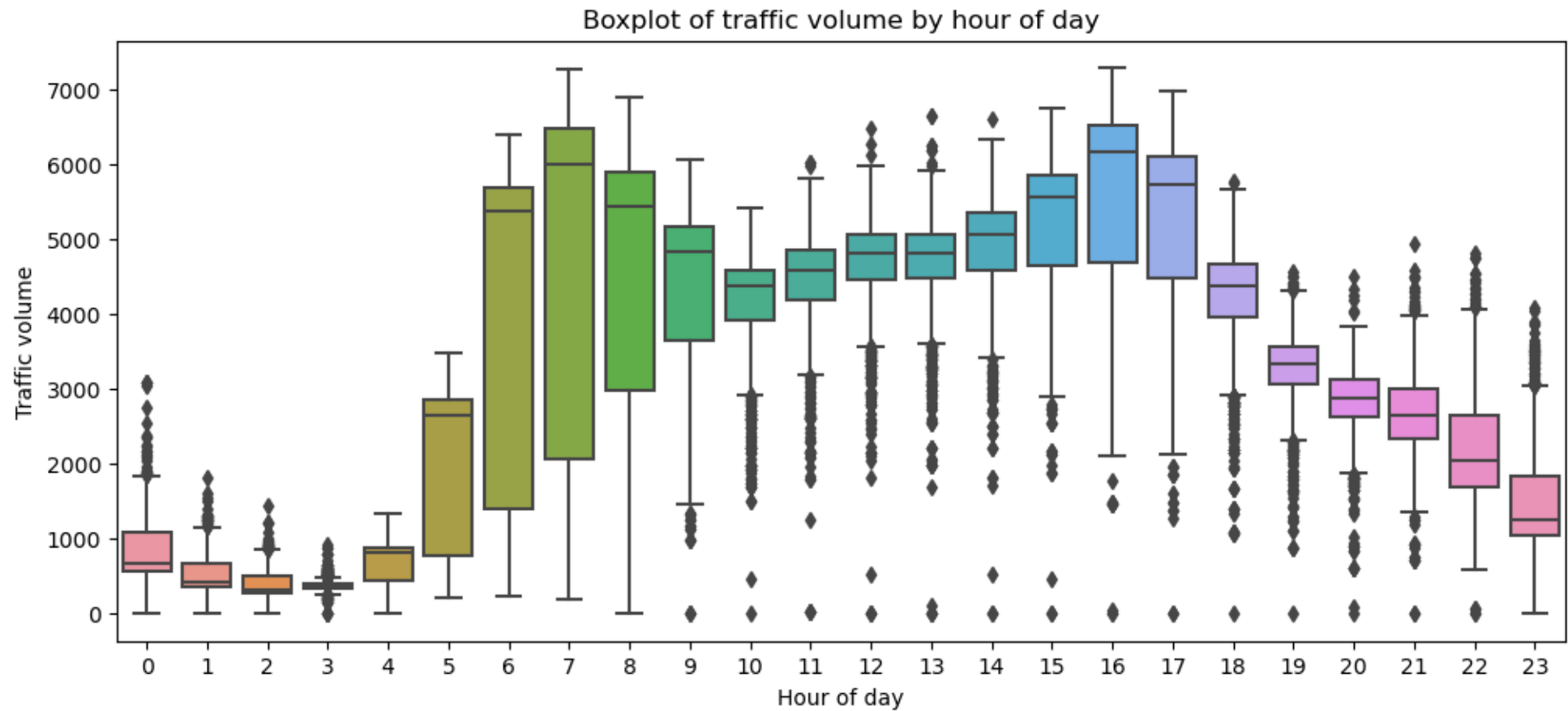
(ATR Dataset)



(ERA5 Dataset)

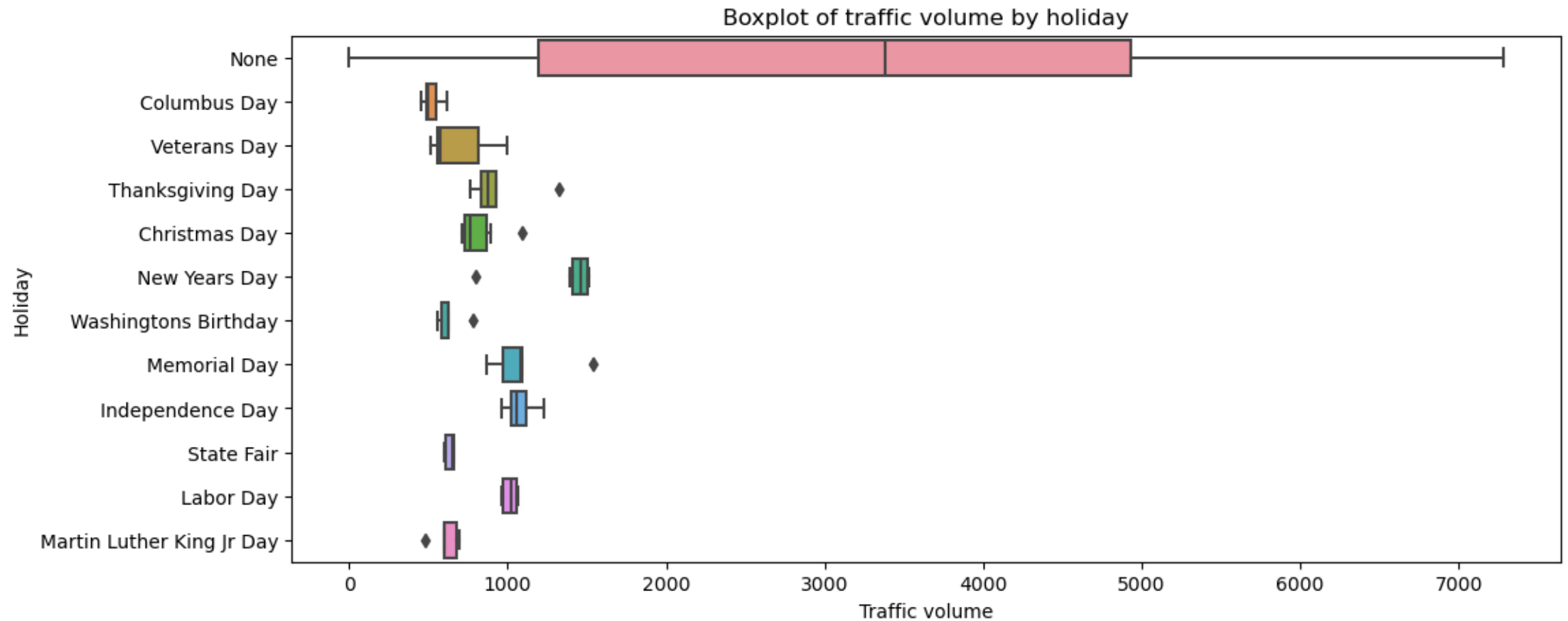
Data exploration

Traffic volume vs. Hour of Day



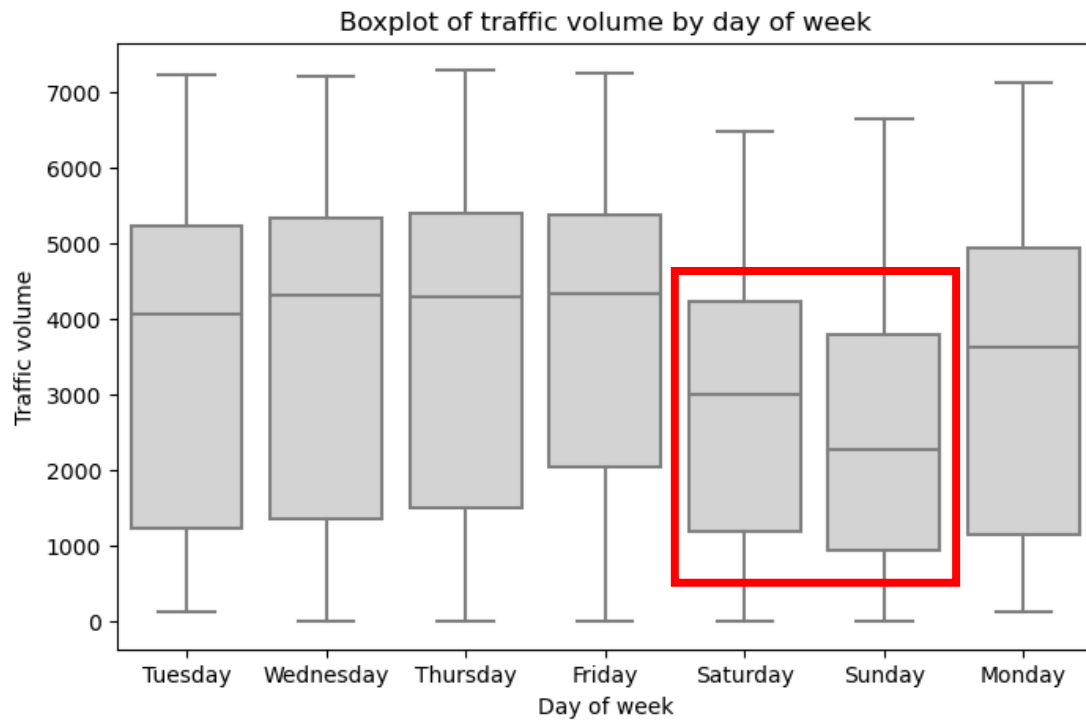
Data exploration

Traffic volume vs. Holidays



Data exploration

Traffic volume vs. Day of Week

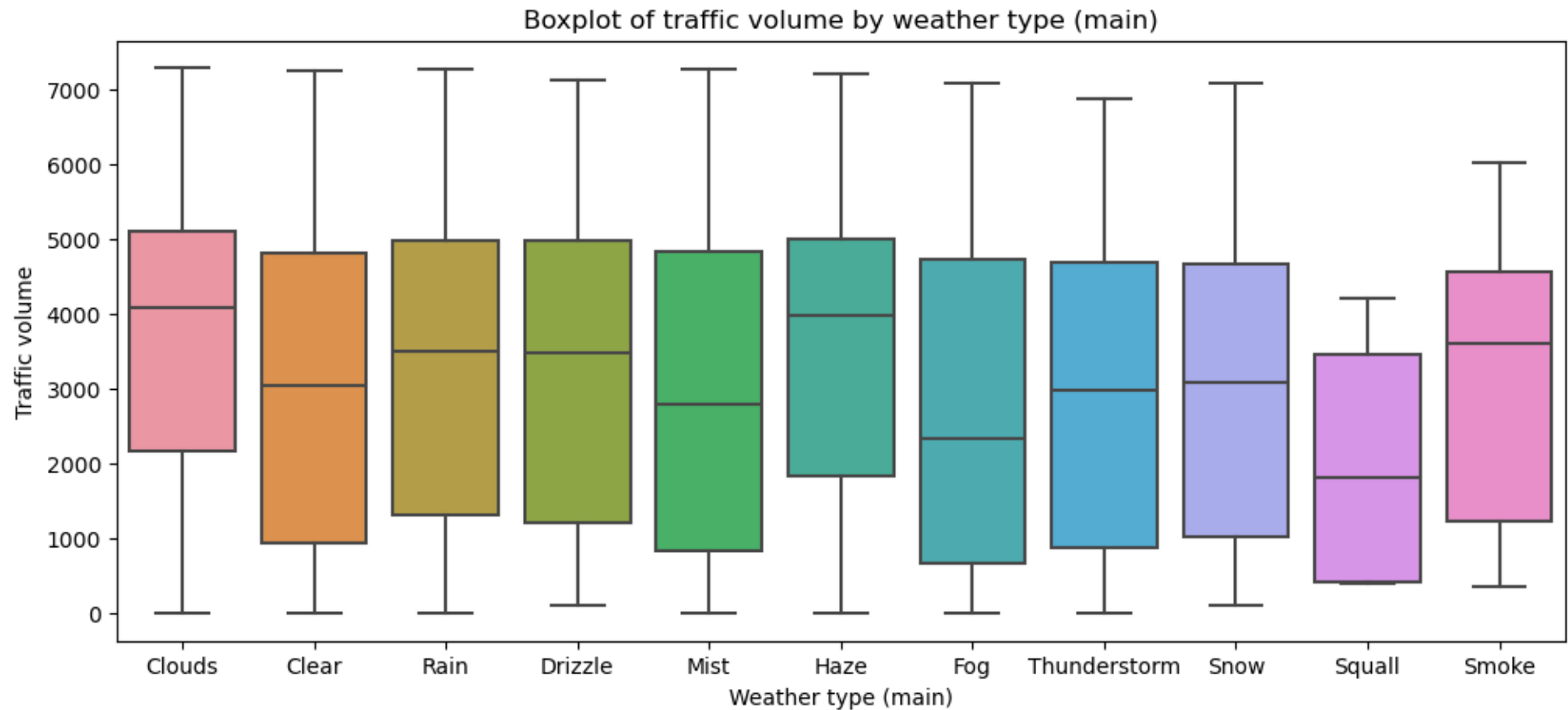


Traffic volume vs. Year



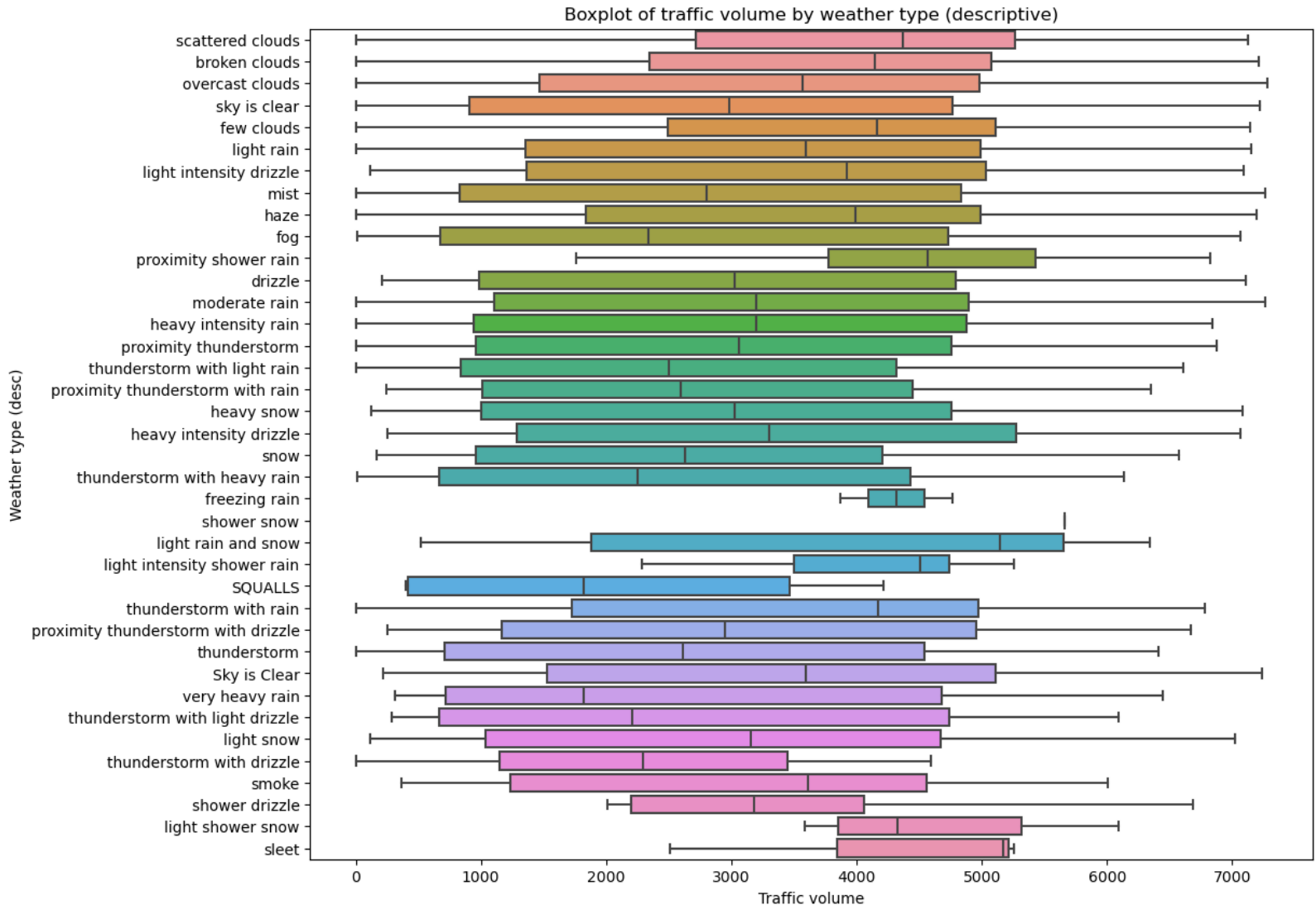
Data exploration

Traffic volume vs. Weather type (ATR, Main)



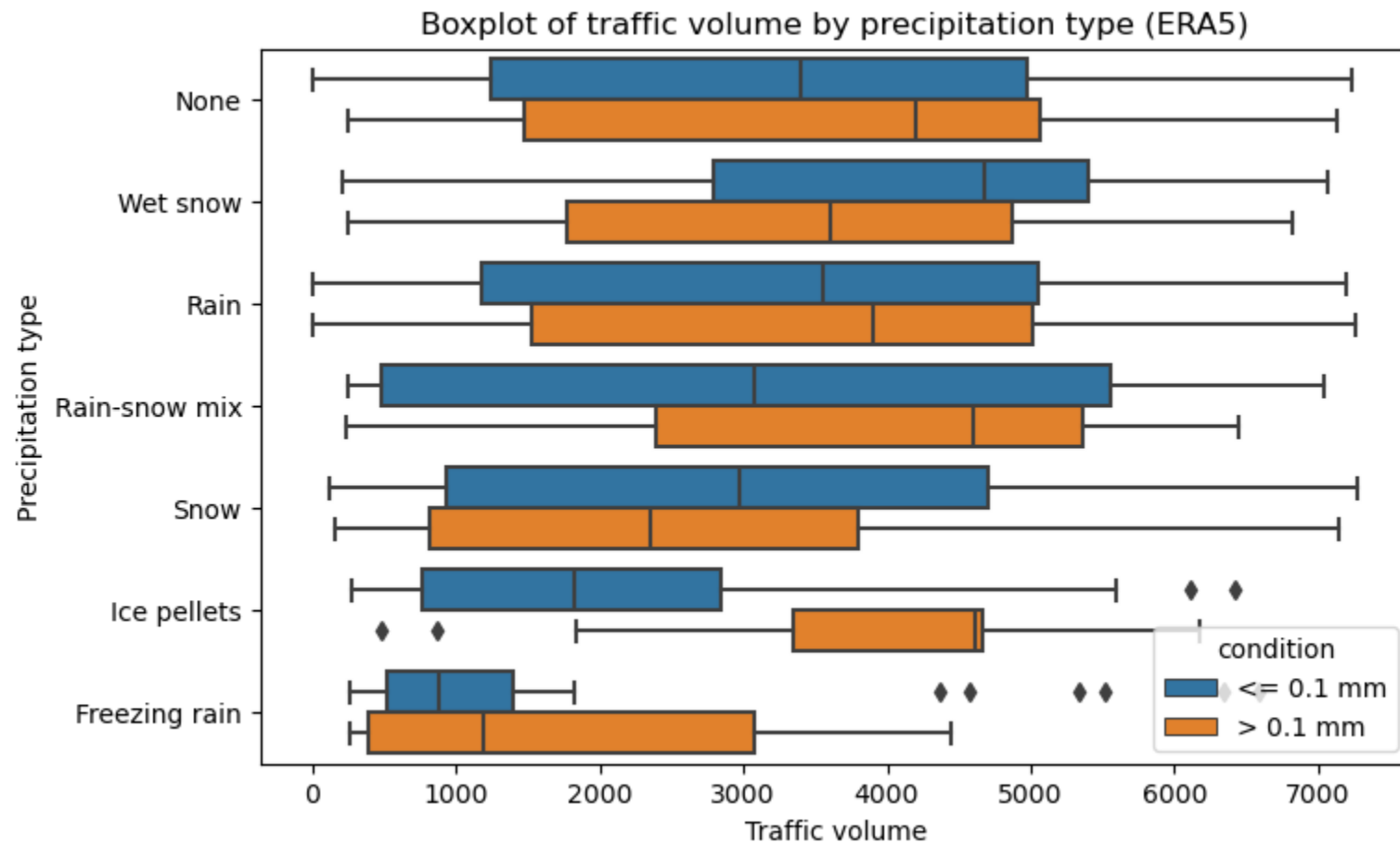
Data exploration

Traffic volume vs.
Weather type
(ATR, Descriptive)



Data exploration

Traffic volume vs. Precipitation type (ERA5)



Data exploration/Conclusion

- Traffic volume is mostly correlated with hour of day
- Expect low correlation from temperature
- Holidays, weekends and mostly importantly, weather types from both datasets DO impact traffic volume!
- **Overall traffic does not change throughout the years,** Meaning we can “simulate” prediction of future traffic!

Model selection

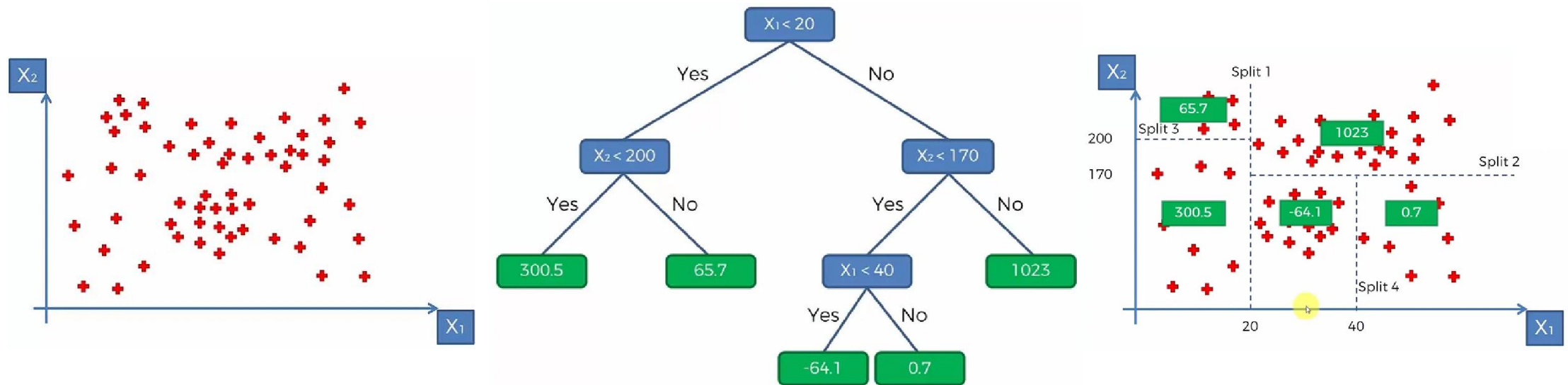
- The non-weather predictor variables are:
 - Temperature (Continuous)
 - Hour of day (24)
 - Day of week (7)
 - Holidays (11)
- For weather predictor variables:
 - **Case 1:** None (0)
 - **Case 2:** ATR, Main (11)
 - **Case 3:** ATR, Detailed (38)
 - **Case 4:** ERA5, Separated by 0.1 mm threshold (14)
- Categorical variables are encoded via one hot encoding.

Model selection

- For regression methods, we'll compare
 - Linear regression
 - Decision tree
 - Random forest
 - Bagging
 - XGBoost
- Train:test data ratio is 7:3.
 - To simulate future traffic prediction,
test data is last 30% by date and time

Model selection

- What is the decision tree?

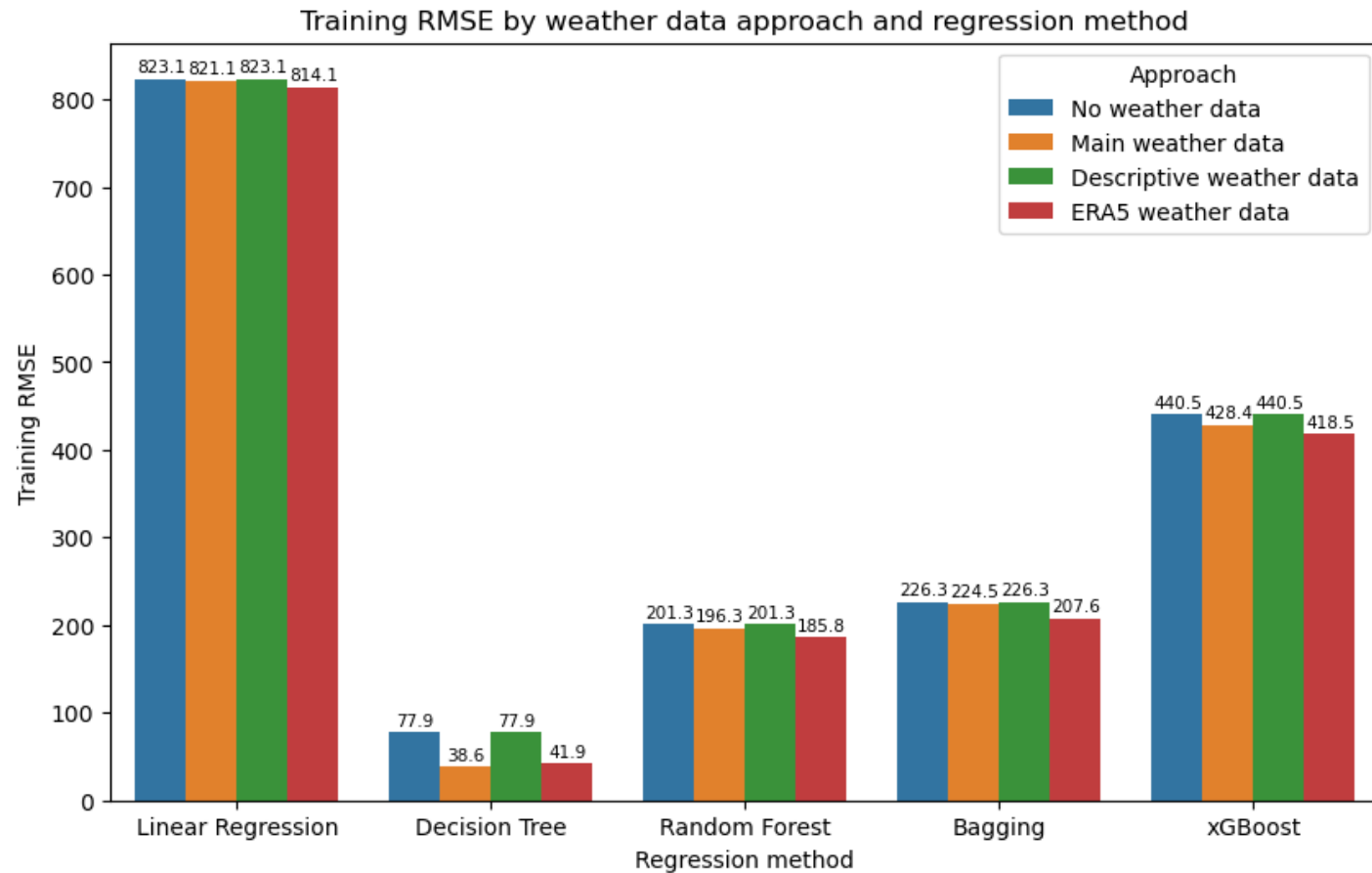


<https://maniksonituts.medium.com/what-is-decision-tree-regression-dcd0ea40a323>

- Random forest, Bagging, and XGBoost methods sample from the train data and evaluate *multiple* decision trees in some way to reduce overfitting.

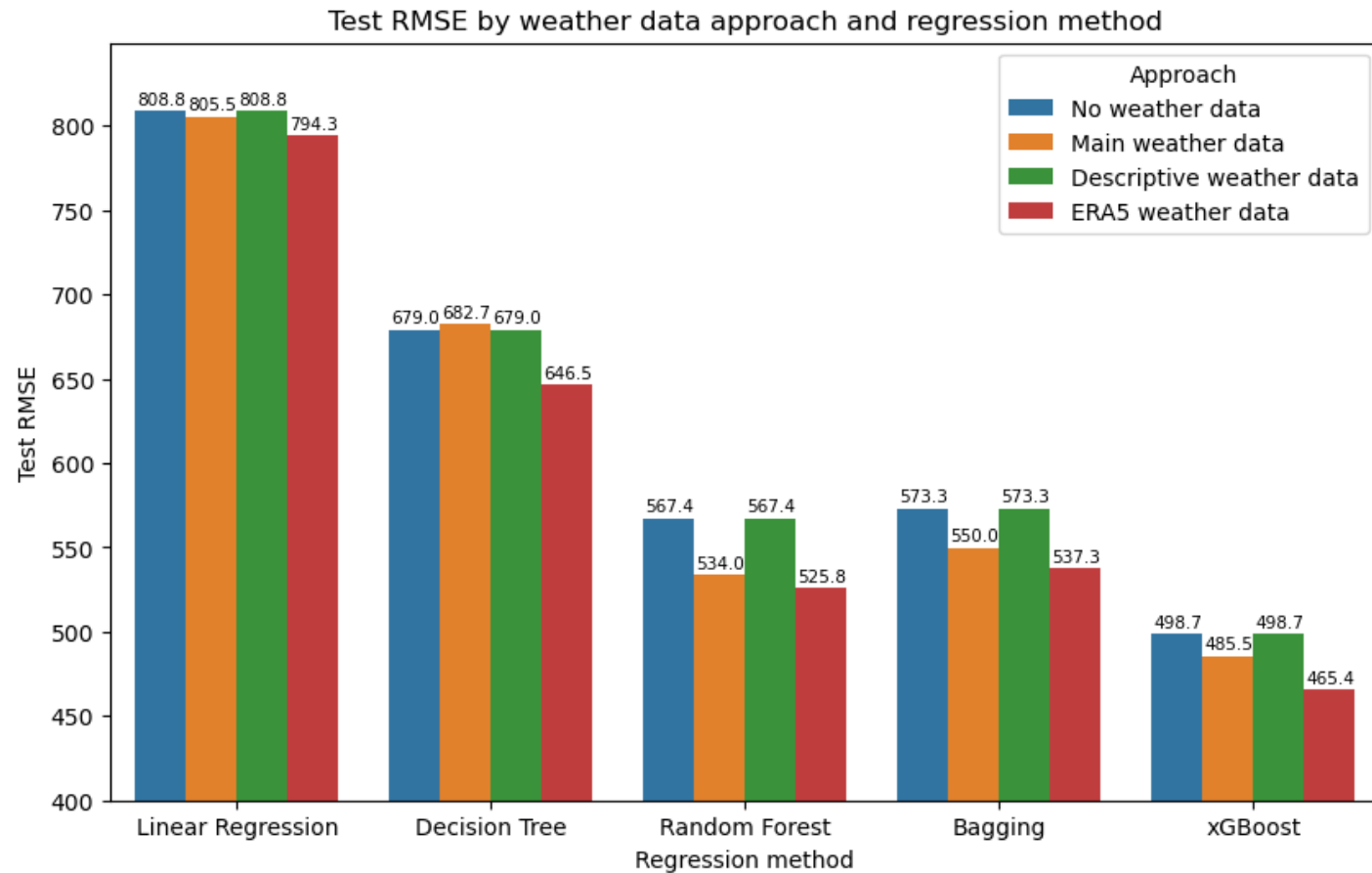
Results

- Train RMSE



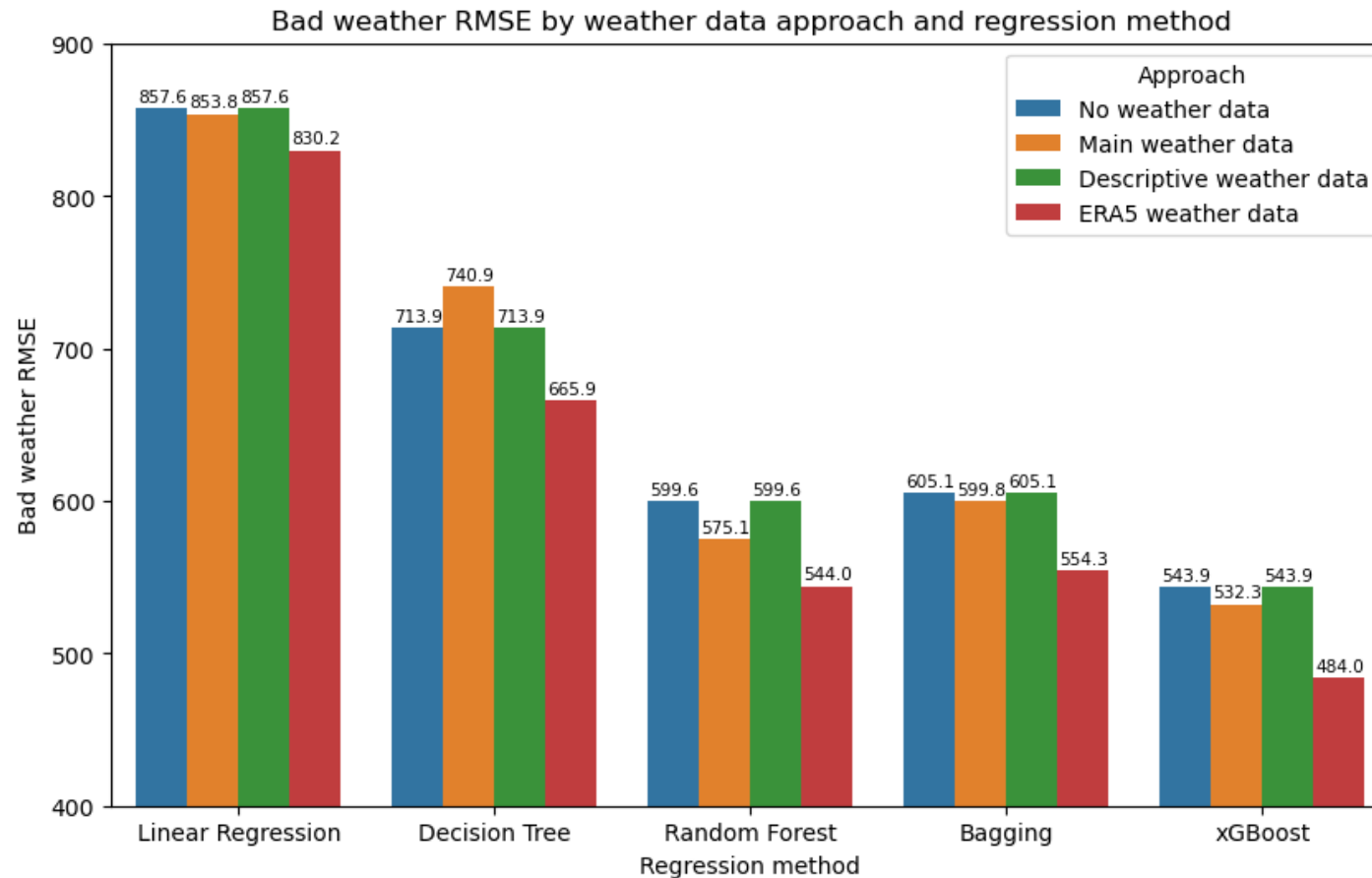
Results

- Test RMSE



Results

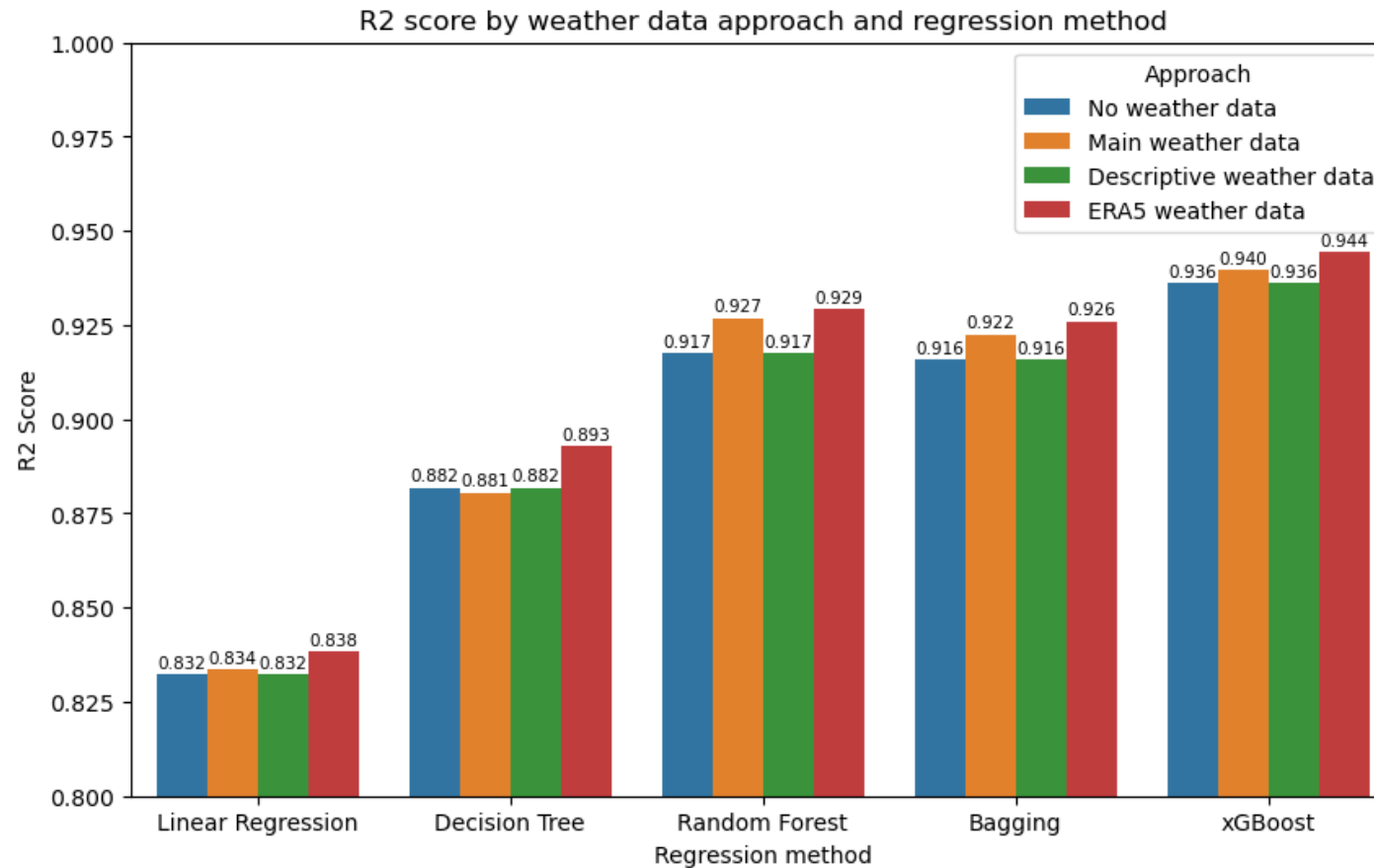
- Test RMSE (Bad weather* only)



*Data where weather is NOT “Clear” or “Clouds” in ATR

Results

- Test R^2



Discussion

- XGBoost showed best goodness of fit, while linear regression showed worst goodness of fit.
- Decision tree, Random forest, and Bagging methods clearly overfit the training data. This is due to not having set the hyperparameters.
- Unexpectedly, both of the local weather measurement data did not improve the test RMSE of the model at all.
- ERA5 reanalysis data consistently improved performance across all regression methods, showing up to 6% in test RMSE overall, and 11% in test RMSE during bad weather.

Conclusion

- This project aims to predict traffic volume based on atmospheric reanalysis data, and compare the performance with local weather measurements.
- Weather type has a significant impact on the traffic volume, and models using ERA5 data outperforms local measurements by up to 6% in test RMSE overall, and 11% in test RMSE during bad weather.
- Atmospheric reanalysis data is a feasible substitute for local weather measurements when building prediction models that require local weather data.

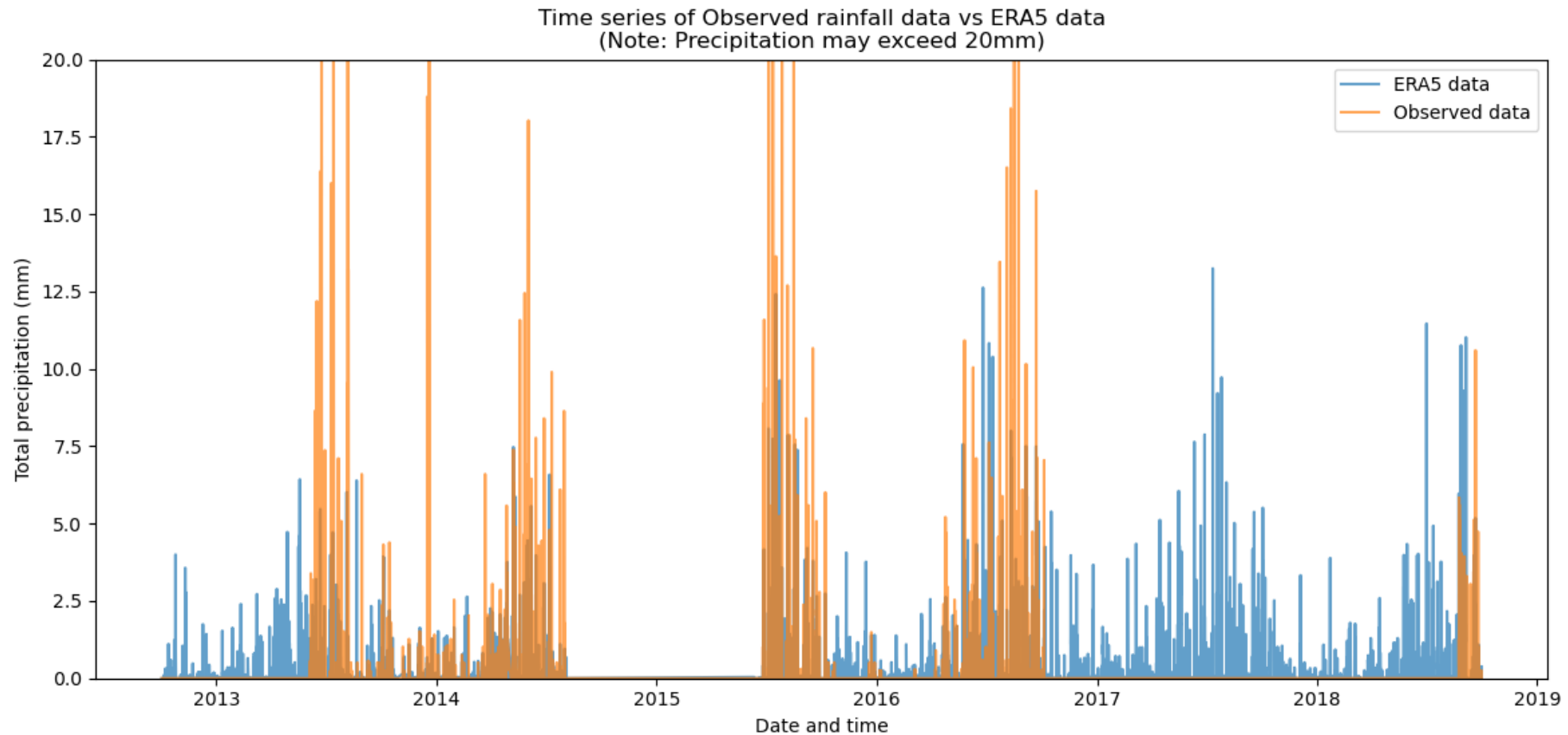
References

- Soni, M. (2020, September 26). What is Decision Tree Regression? *Medium*. <https://maniksonituts.medium.com/what-is-decision-tree-regression-dcd0ea40a323>

Thank You for Listening!

Appendix

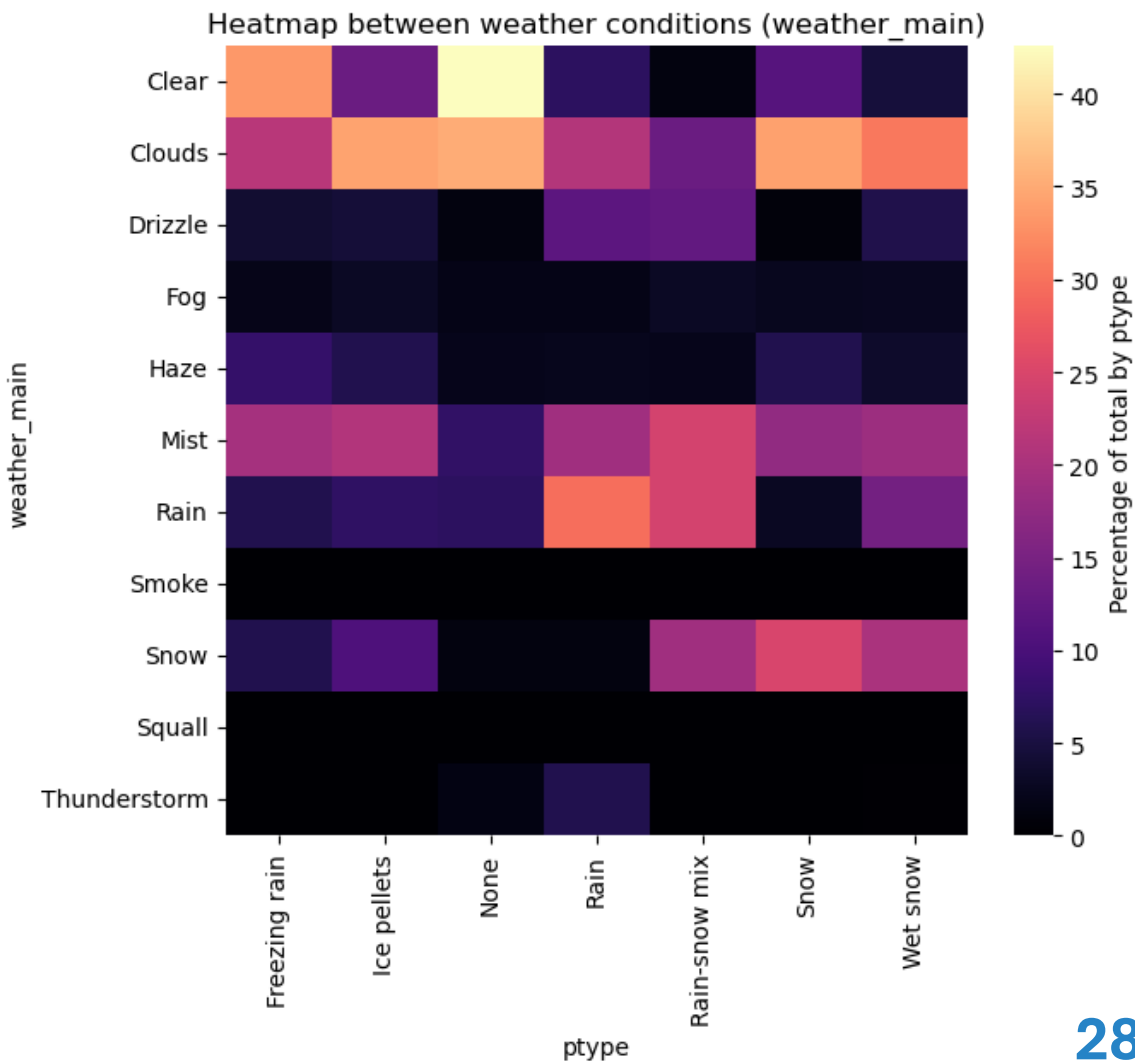
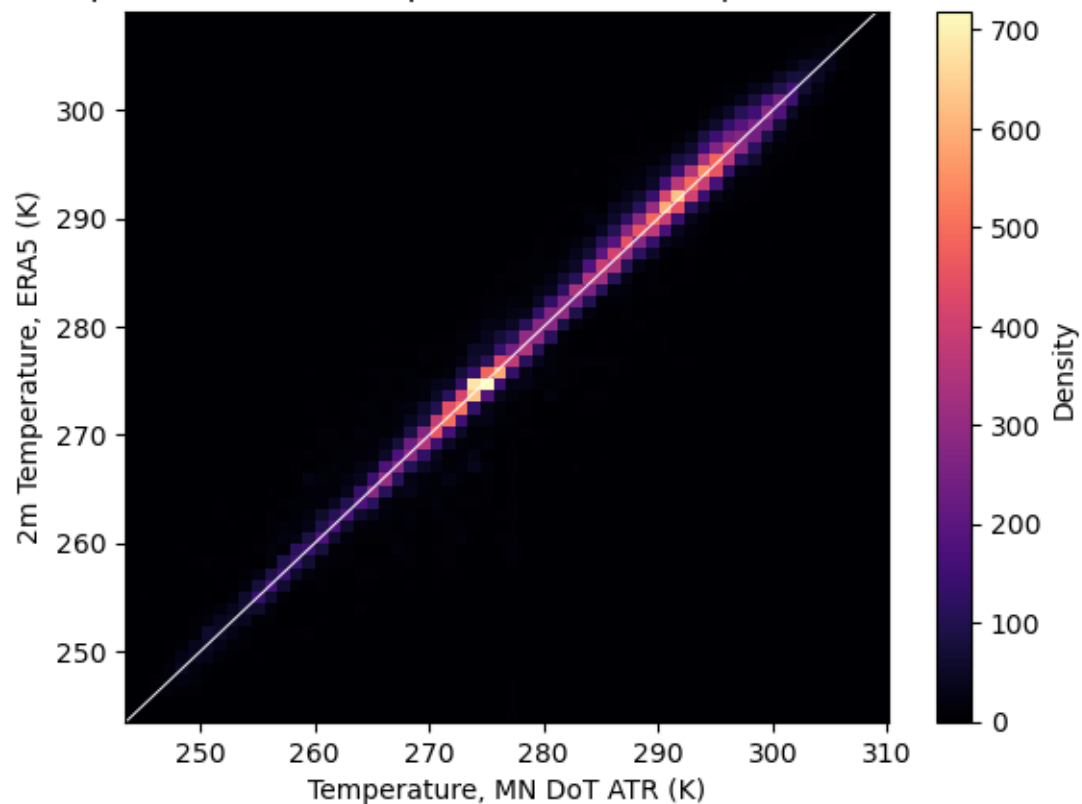
- Time series of total precipitation



Appendix

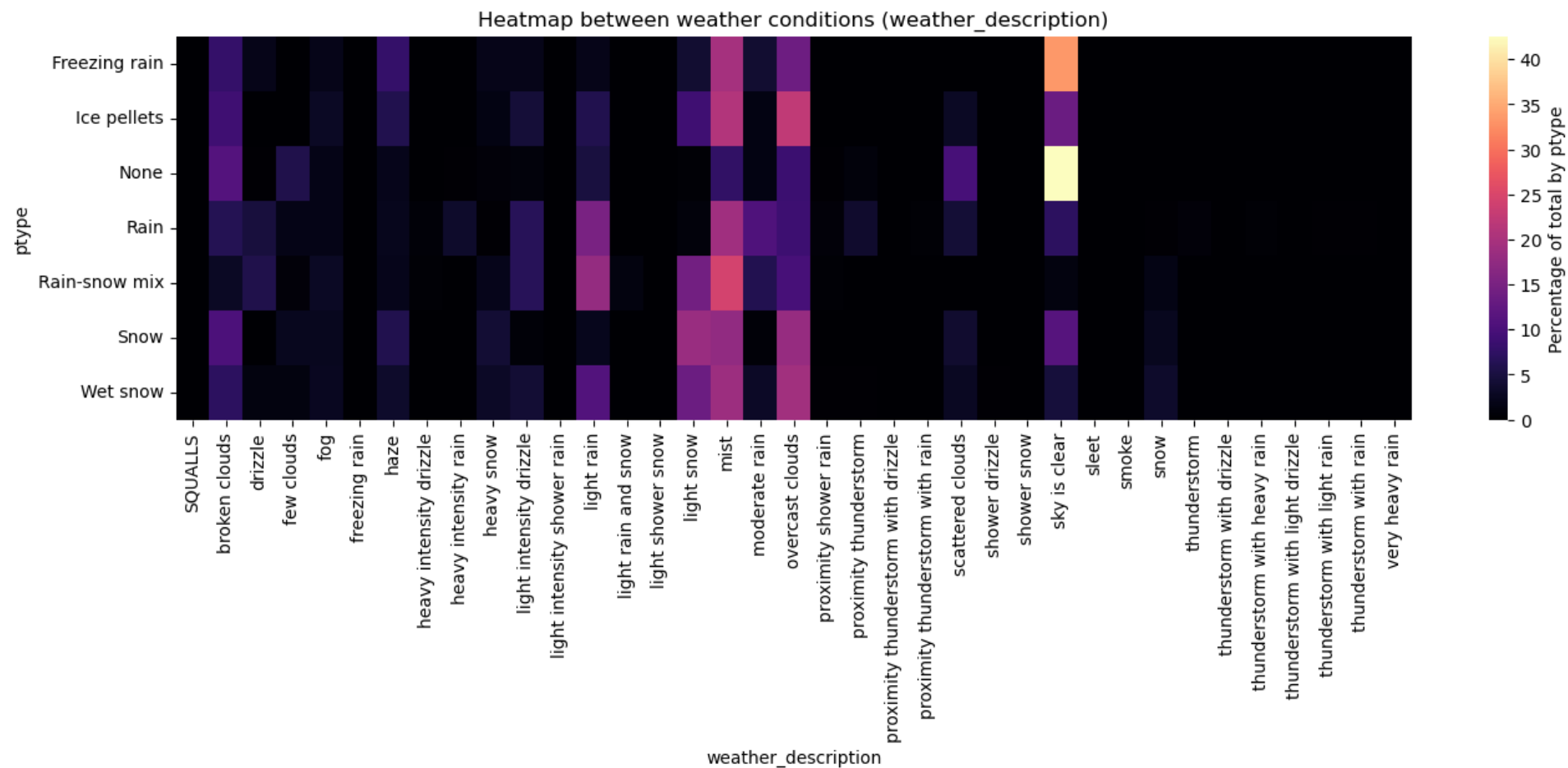
- Data correlation by datasets

Temperature relationships between two acquisition methods



Appendix

- Data correlation by datasets



Appendix

- Locating coordinates of ATR data

Roadway Project Mapping Application
HPMS Year 2023

Layers

Identify

Find

Measurement

Find Segment on Map

Find Carto Mile on Map

Find Reference Post on Map

Enter EITHER a Route Name OR Route ID and reference post + offset in the correct format as shown:

Route Name: I 94 I

Route ID: 0100000000000094-I

Ref Post: 239+00.922

Locate

Clear Results

Result

Status: OK
Route ID: 0100000000000094-I
Carto Mile: 240.580

Route Info Tool

MONTHLY REPORT, STATION NO. 301 (ATR)
JANUARY 2018

Station Measurement: VOLUME		
Route: I-94	Route Direction: E/W	Lanes: 8
County: Ramsey	Closest City: St Paul	
Functional Class: Urban Principal Arterial - Interstate		
Location: W OF N VICTORIA ST IN ST PAUL		
Ref Post: 239+00.922	True Mile: 240.541	Sequence No. 9834
Volume: 4,507,471		MADT: 145,402
Weekday (M-F) MADT: 154,006		Weekend (Sa-Su) MADT: 123,072

https://dot.state.mn.us/traffic/data/reports/atr/Monthly_PDFs/Jan18/301.pdf
<https://mndotgis.dot.state.mn.us/rpma/>

30